Four-mesh Static Circuit

Consider the example circuit in the following.

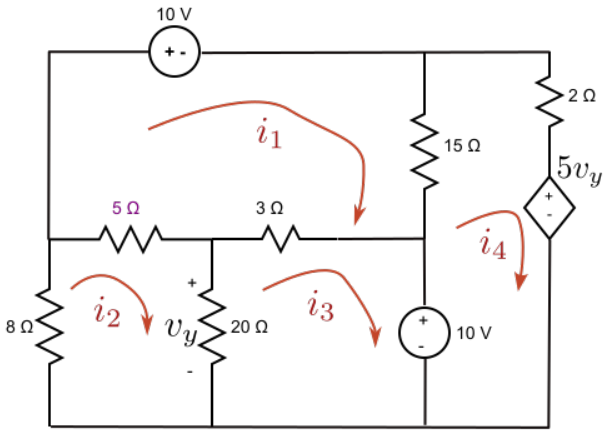


Figure 1: Four-loop Mesh Circuit

In order to find the current in each of the four loops, i_1, i_2, i_3, i_4, the problem can be formulated as a system of linear equations $Ax = y$.

Problem

Write function `solve_fourmesh` to solve the current in the loops for this problem. Declare your loops condition matrix A and vector y from the circuit given. The circuit current vector can be calculated by solving for x such that:

$$Ax = y$$

Implementation

Goal: Fill in the following function then submit your function in grader:

```
import numpy as np

def solve_fourmesh():
    A = ...
    y = ...
    x = ...

    return A, y, x # return calculated currents for each loop
```